

Legislative Coordination and Institutional Change

A Bipartite Network Analysis of Coalition Dynamics in the Brazilian Chamber of Deputies

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| The puzzle

- ◆ Polarization should make legislatures **ungovernable**.
- ◆ Brazil 2014–16: impeachment · Lava Jato · recession · polarization.
- ◆ Yet the Chamber **kept forming majorities** — across impeachment and a change of power.
- ◆ **How? And at what cost?**

Coalitional presidentialism, and the 2014–16 inversion

- ◆ Presidential powers, no majority → govern by **coalition** (Abranches 1988).
- ◆ Glued by budget, posts, agenda; the **Centrão** sells support as *agent*.
- ◆ 2014–16: presidency weakens; budget tool moves to Congress (*impositive*, 2015).
- ◆ Relation **inverts**: the pivot starts to *set* terms (*governo congressional*).

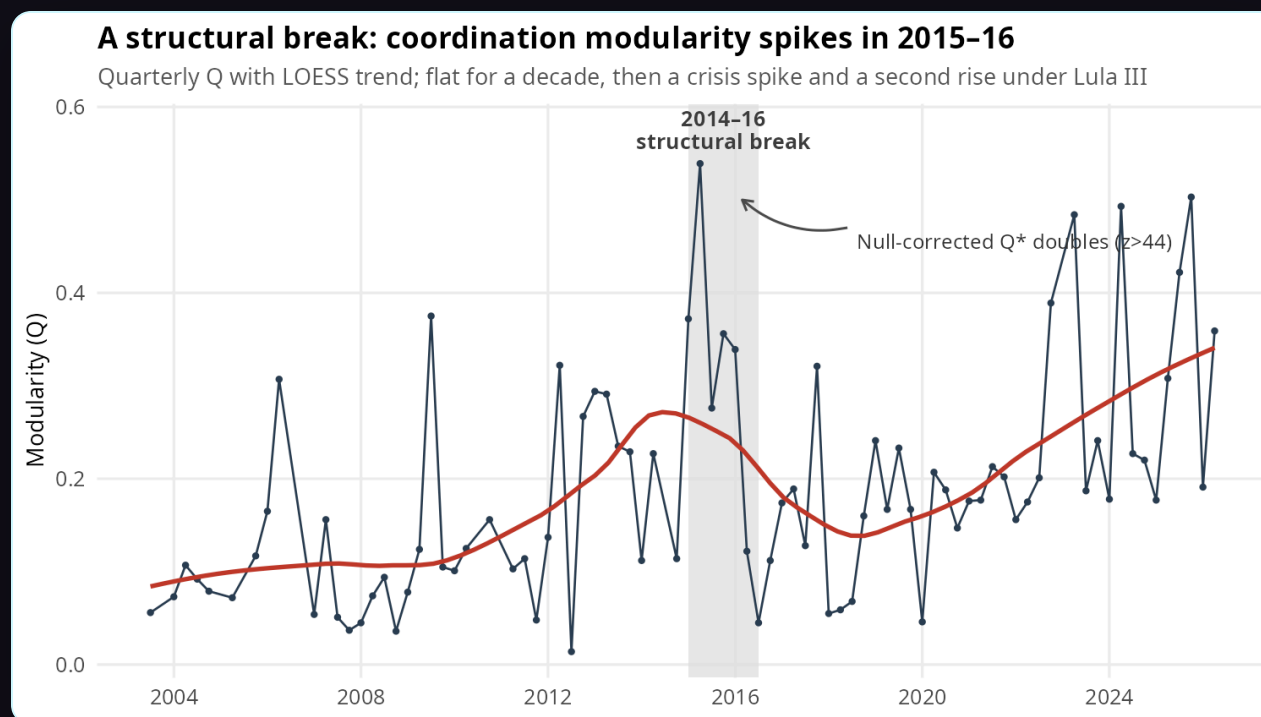
| The thesis, in one line

- ◆ The pragmatic center is now the pivot through which every government builds majorities.
- ◆ Coordination by **position vs. the executive**, not by ideology.
- ◆ The Centrão is **old**; its **position** changed: *agent* → *pivot*.
- ◆ Backdrop: 2015 *impositive* amendments — budget power shifts to Congress.

| Why networks, not ideal points

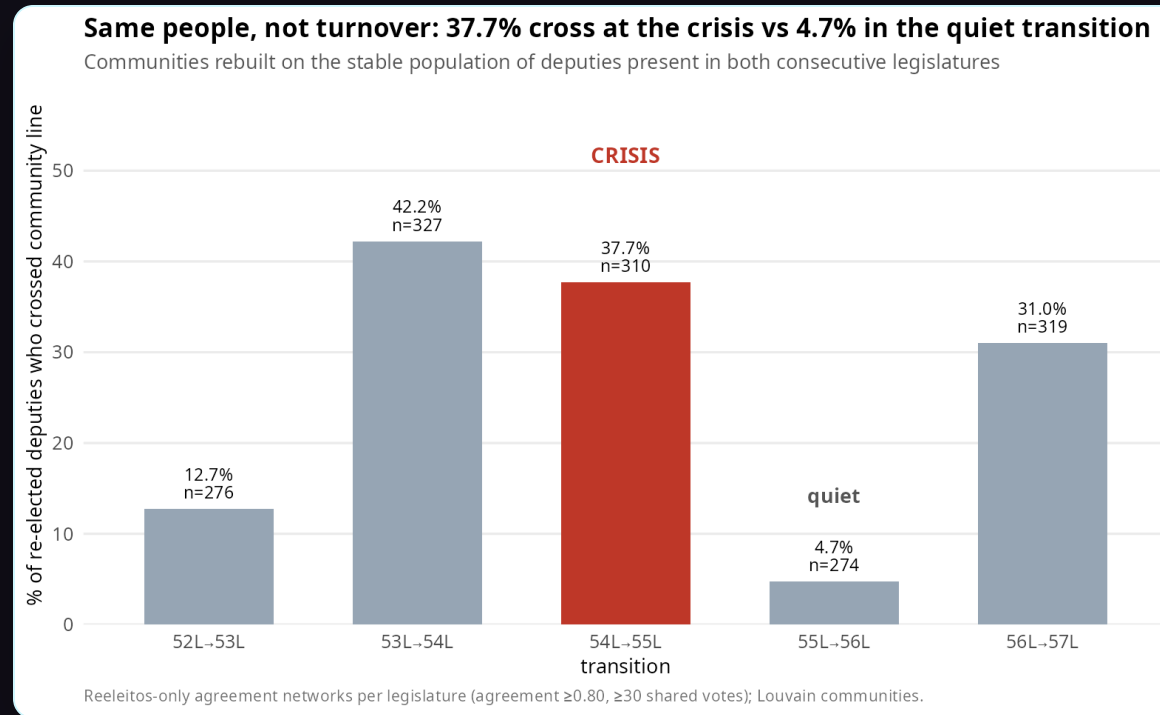
- ◆ **deputies × bills** → agreement graph → **communities**.
- ◆ Ideal points force one **left-right axis**.
- ◆ Blocs **recompose** while ideology looks stable — an axis can't see it.
- ◆ No axial assumption → the shift is **bloc recomposition**.

A structural break: modularity spikes in 2015–16



Quarterly Q: flat ~0.15, spikes to 0.51 in 2016. Null-corrected Q^* doubles at the 55L and 57L.

Same people, not turnover



Reelected deputies crossing communities: **37.7%** at the break vs **4.7%** after.

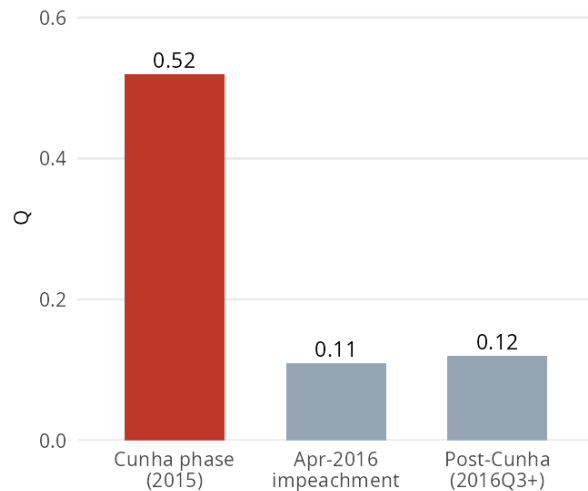
Not turnover. Not agenda. Not just fiscal.

The reorganization survives the observable rivals

Collinear shocks preclude estimating one effect; the eliminative design kills each rival in turn.

Not agenda control

Recovery dates to the impeachment, not to Cunha's exit



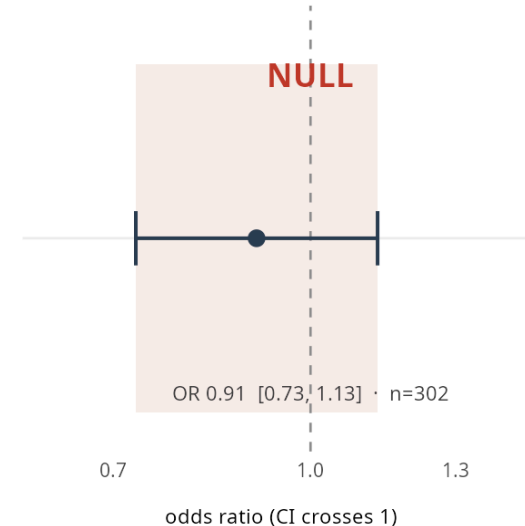
The break is real

Per-legislature Q^* , highest at 55L and 57L ($z > 44$)



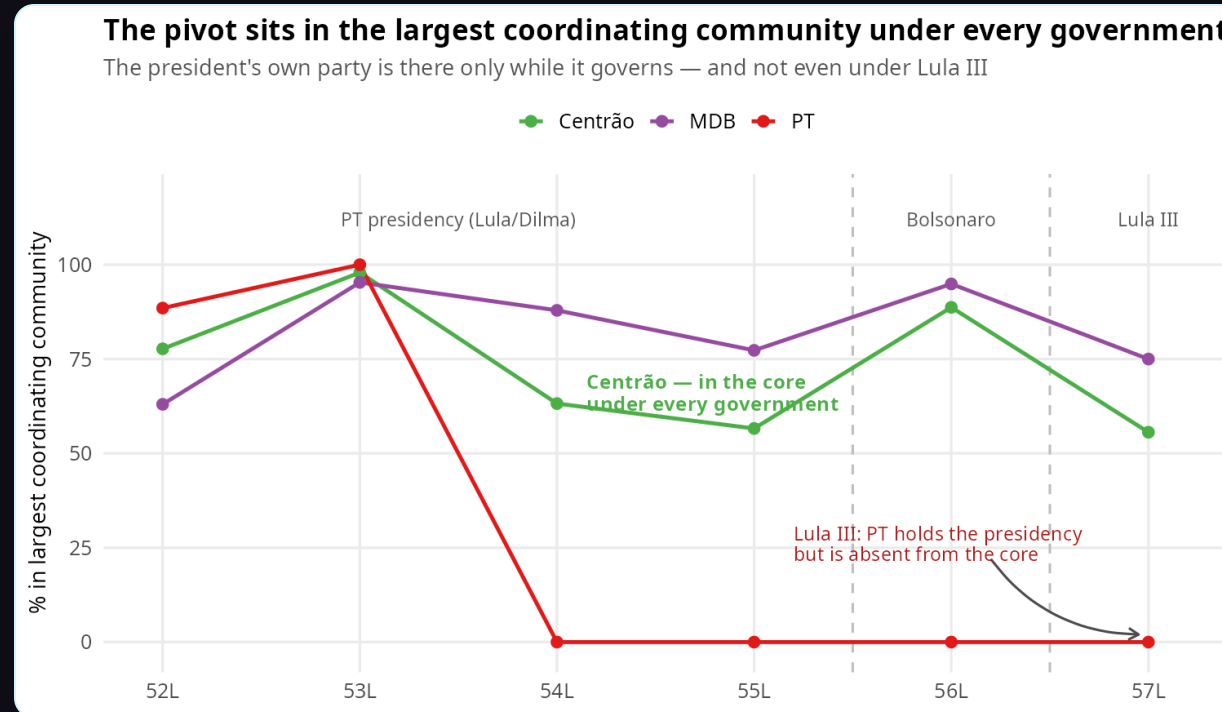
Not fiscal-specific

Amendment exposure doesn't predict crossing



The break survives every rival. Shocks are collinear → the design eliminates rivals, not estimates one effect.

The pivot: in the largest community under every government



Centrão present under every government; the president's party only while it governs — not even under Lula III.

| From agent to principal, and the cost

- ◆ Budget transfers **reverse the relation**: the pivot **sets the terms** (*governo congressual*).
- ◆ Governability comes to be **sold, not represented**.
- ◆ Cost = **accountability**, not gridlock: a bloc available to any government is hard to hold to account.

| Contribution, and what the case teaches

- ◆ **Measurement:** a network quantity for the pivot's position, persistence, decision weight.
- ◆ **Design:** an eliminative chain for **collinear shocks**.
- ◆ **Not only Brazil:** cousins in Israel, post-war Italy — change **without collapse**.

A positional realignment the standard typologies miss: reorganization without collapse.

Thank you

Questions welcome. Backup slides follow.

Andéliton Soares

Method

- ◆ Pairwise Krehbiel-style agreement on co-voted bills; edge filter on agreement and shared-vote count.
- ◆ Community detection: **Louvain** (Leiden interchangeable, $ARI \approx 0.98-1.00$).
- ◆ Q^* : observed modularity minus mean of a within-votation permutation null (150–200 draws); $z > 44$, $p < 0.001$.
- ◆ Within-deputy panel: reelected members across consecutive legislatures, matched by majority overlap.

Robustness

- ◆ **Thresholds:** 12-cell grid (edge 0.70–0.85 × shared-vote 20–40); 54L → 55L crossing in [29.6%, 44.8%].
- ◆ **Projection:** Jaccard and resource-allocation robust; collapses only under Newman (no community structure).
- ◆ **Algorithms:** Louvain, Leiden, Infomap, Walktrap; crisis transition least similar on label-free ARI/NMI.
- ◆ **Near-unanimous votes** dropped (modal > 95/90/85%): crossing holds at 30.6 / 28.6 / 26.0%.

| Why not a temporal ERGM

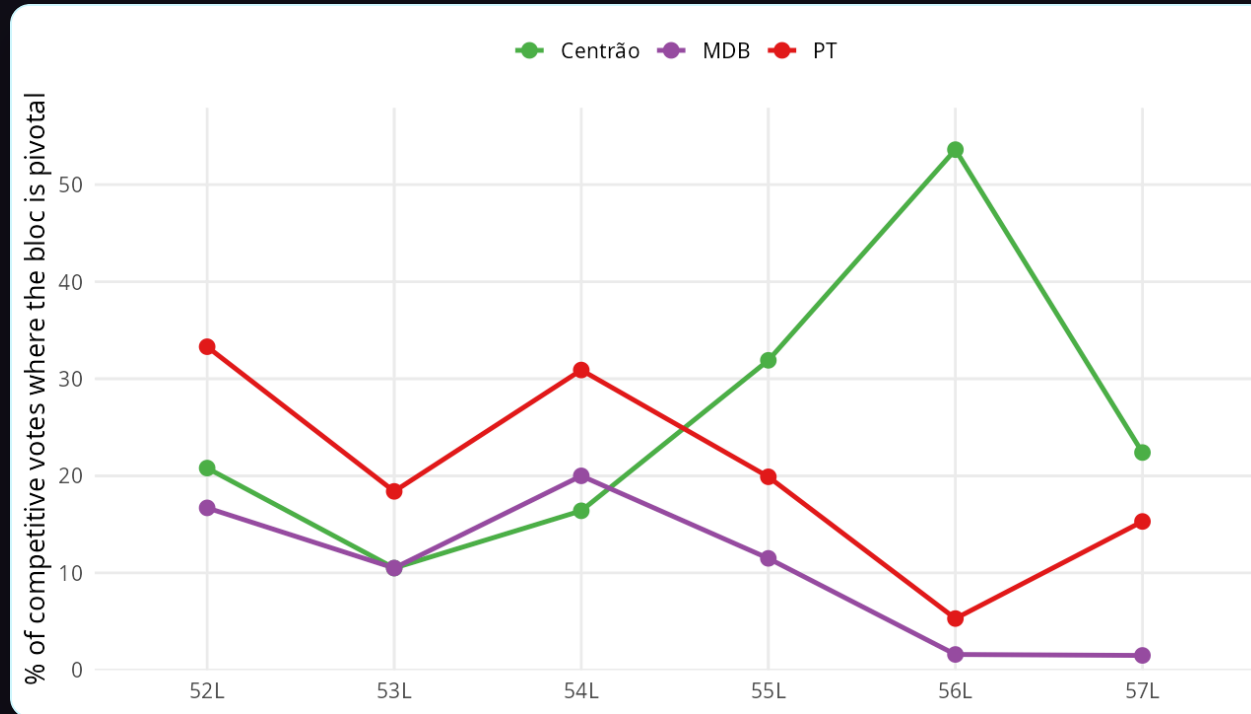
- ◆ Given collinear shocks and poor mixing in the bipartite specification, a bootstrap-TERGM with the amendment reform as covariate did **not provide a reliable route to the estimand** here.
- ◆ OLS-DiD: the reform's coefficient on modularity **collapses** under controls; variance absorbed by the Cunha dummy.
- ◆ Static bipartite ERGM on the 54L did **not mix** after 137 minutes (estimation degenerated).
- ◆ Hence the eliminative design: an **informative outcome**, not a hidden failure.

| The Centrão, defined

BACKUP

- ◆ A reproducible, sourced, **dated** set of mid-sized, low-programmatic, *governista* parties (Testa et al. 2024).
- ◆ Each party enters only from the legislature it **joined the governista orbit** (e.g., DEM from the 55th).
- ◆ PSB coded left (out); MDB tracked separately as a pivot.
- ◆ Survives **six** membership schemes; the bloc grew from **19.5%** to **66.1%** of the chamber.

Decision weight migrates from PT to Centrão



On competitive roll-calls, the decisive core migrates from the PT (while it governs) to the Centrão (net of the president's party).